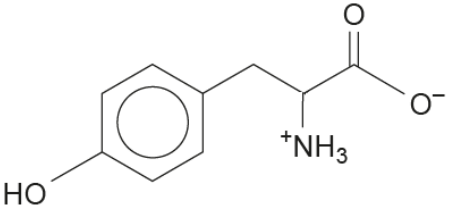


Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
11	(a)	(i)		chromatogram drawn correctly with spot at 6 cm mark		1		1		1
		(ii)			1			1		
		(iii)		the polar structure / OH group is a small part of the overall molecule so hydrogen bonding is at a 'minimum'			1	1		
		(iv)	I	24500 cm³ of nitrogen from 181 g of tyrosine (1) 1 cm³ of nitrogen from $\frac{181}{24500}$ g of tyrosine 147 cm³ of nitrogen from $147 \times \frac{181}{24500} = 1.09 \text{ g}(1)$ accept alternative methods e.g. $n = \frac{pV}{RT} = 0.006$ (1) mass = $0.006 \times 181 = 1.09\text{g}$ (1) e.g. 1 mol tyrosine gives 1 mol N₂ $n(\text{N}_2) = \frac{1.09}{181} = 6 \times 10^{-3}$ (1) $V = \frac{nRT}{p} = 147\text{cm}^3$ (1)		2		2	1	

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
			II	award (1) for any of following not all the nitrogen was collected impure sample of tyrosine erroneous starting mass incomplete reaction			1	1		1
	(b)	(i)		M_r of calcium butane-1,4-dioate = 156 (1) atom economy = $\frac{156}{(74 + 180)} \times 100 = 61$ (1)	1	1		2		
		(ii)		moles of calcium butane-1,4-dioate = $\frac{41.2}{156} = 0.264$ moles of H_2SO_4 needed = 0.264 (1) volume of H_2SO_4 needed = $\frac{0.264 \times 1000}{2.5} = 106 \text{ cm}^3$ (1)		2		2	1	
	(c)			H H₃C C=C C=C H H H H C=CH₃ C=CH there are 3 peaks (1) award (1) for recognising the equivalence of three pairs of C atoms can be labelled on structure or in statement e.g. both CH₃ carbon atoms are equivalent, both 'end' carbon atoms of double bonds are equivalent and both 'internal' carbon atoms of double bonds are equivalent		1	1	2		
				Question 11 total	2	7	3	12	2	2